

Systems and Methods for Autonomous, Threat-Intelligence-Driven Deployment and Behaviorally-Adaptive Configuration of Deception Assets

Inventor: Harshavardhan Malla | Applicant: NovusAI

Provisional Patent Application (Draft)

Contents

1 Systems and Methods for Autonomous, Threat-Intelligence-Driven Deployment and Behaviorally-Adaptive Configuration of Deception Assets	1
1.1 Cross-Reference to Related Applications	1
1.2 Technical Field	1
1.3 Background	2
1.4 Summary of the Invention	3
1.5 Brief Description of the Drawings	3
1.6 Detailed Description	4
1.7 Claims	7
1.8 Abstract	9

1 Systems and Methods for Autonomous, Threat-Intelligence-Driven Deployment and Behaviorally-Adaptive Configuration of Deception Assets

Application type: Provisional patent application (draft for filing) **Inventor:** Harshavardhan Malla **Applicant / Assignee:** NovusAI **Priority date:** to be assigned at filing

1.1 Cross-Reference to Related Applications

This application is associated with the Novus Aegis AI dynamic-honeypot program documentation (v1.0, 11 February 2026) and with the inventor’s research on behavioral-cognitive attacker-intent modeling from honeypot interaction telemetry. No related applications are presently on file.

1.2 Technical Field

The present invention relates to network security, cyber-deception, and autonomic computing. More particularly, it relates to a system and method for autonomously deploying, configuring, randomizing, and continuously adapting a population of deception assets (honeypots and related

decoys) across one or more computing environments, wherein deployment and configuration decisions are driven by live threat intelligence and by inferred attacker intent derived from observed interaction behavior, and wherein all autonomous actions are bounded by machine-enforceable cost, exposure, and compliance policies.

1.3 Background

1.3.1 Deception and its operational value

A honeypot is a decoy computing resource instrumented to attract and observe unauthorized interaction. Because legitimate users have no reason to touch a decoy, interaction with a honeypot is inherently high-signal: it yields low-false-positive detections and rich forensic evidence of adversary tactics, techniques, and procedures (TTPs). Deception is therefore attractive for security operations centers (SOCs), threat hunting, and incident response.

1.3.2 Limitations of the prior art

Despite this value, conventional deception suffers from systemic limitations that the present invention addresses.

1. **Static configuration and fingerprintability.** Conventional honeypots are configured once and remain fixed. Their service banners, open-port sets, operating-system fingerprints, response timing, and emulated vulnerabilities do not change. Sophisticated adversaries and automated reconnaissance frameworks maintain signatures of common decoy software and known-default deployments, and can therefore fingerprint and avoid static honeypots. Once a decoy is recognized, its intelligence value collapses.
2. **Slow adaptation to a shifting threat environment.** Adversary campaigns shift rapidly across targeted services, vulnerabilities, and regions. A statically placed decoy does not track current threat intelligence and consequently provides poor coverage of emerging campaigns and presents a low-value target at the wrong place and time.
3. **Manual operational overhead that does not scale.** Selecting where, when, and how to deploy decoys, and re-tuning them over time, requires scarce expert labor. This does not scale across multi-cloud, hybrid, and multi-tenant estates.
4. **Low-fidelity, low-context output.** Static honeypots emit large volumes of raw logs rather than prioritized, enriched, attributable intelligence. Analysts must manually correlate sessions into actors and campaigns.
5. **Absence of intent modeling.** Prior deception treats all interaction uniformly and reacts only after an attacker has acted (for example, after a payload is delivered). It does not infer, from early pre-action behavior, *what the attacker is trying to do*, and therefore cannot adapt the decoy in time to maximize engagement and intelligence yield.
6. **Unbounded or unsafe autonomy.** Approaches that introduce automation rarely bound it: there is no principled, machine-enforceable control of cost, public exposure, region, data handling, and compliance, nor an auditable record of why each autonomous action was taken.

There is accordingly a need for a deception system that (a) closes the loop between live threat intelligence, autonomous deployment, observation, and learning; (b) continuously randomizes the

deception surface to resist fingerprinting; (c) infers attacker intent from early behavior and adapts accordingly; and (d) constrains all autonomy with explainable, machine-enforceable policy.

1.4 Summary of the Invention

The invention provides a closed-loop, policy-bounded, AI-driven deception system. In various aspects:

- An **artificial-intelligence decision engine** ingests normalized threat intelligence and internal telemetry and, subject to machine-enforceable policy, emits an **explainable deployment instruction** specifying a honeypot type, location, interaction level, emulated vulnerability profile, and time-to-live, together with a rationale identifying the contributing signals.
- An **orchestrator** provisions the specified deception assets across one or more computing environments using infrastructure-as-code, and a **dynamic configuration manager** configures them and **randomizes their deception surface** so that no two deployments present an identical fingerprint, thereby resisting adversarial reconnaissance.
- A **behavioral-cognitive intent model** infers, from early pre-action interaction features, the probable intent or archetype of an interacting adversary, and the decision engine uses the inferred intent to adapt the configuration of the engaged decoy and the placement of subsequent decoys.
- A **monitoring and correlation subsystem** converts observed interaction into enriched, technique-mapped alerts and into longitudinal attacker profiles, and returns observed behavior to the decision engine, **closing a feedback loop** that continuously optimizes the deception population.
- A **policy integrator** enforces cost, exposure, region, retention, and compliance constraints at both decision time and execution time, so that autonomy is bounded and auditable.

These aspects may be practiced separately or in combination, and the deception surface randomization and the intent-driven adaptation each constitute independent inventive concepts.

1.5 Brief Description of the Drawings

- **FIG. 1** — System block diagram of the closed-loop deception platform.
- **FIG. 2** — Flow diagram of the end-to-end loop: ingest → decide → deploy → configure-and-randomize → monitor → infer-intent → alert/correlate → feed back.
- **FIG. 3** — The deception-surface randomization process across multiple fingerprint dimensions.
- **FIG. 4** — The behavioral-cognitive intent-inference pipeline operating on early, pre-action interaction features.
- **FIG. 5** — Policy-guardrail enforcement at decision time and execution time.
- **FIG. 6** — Conceptual data model of the principal domain objects.
- **FIG. 7** — A multi-cloud, multi-tenant deployment topology with time-to-live recycling.

1.6 Detailed Description

1.6.1 1. System overview

Referring to FIG. 1, the system comprises cooperating modules forming a closed control loop over a population of deception assets: a threat-intelligence ingestion module; an artificial-intelligence decision engine; a honeypot orchestrator; a dynamic configuration manager; a behavioral-cognitive intent model; a real-time monitoring and alert subsystem; a threat-intelligence logger and correlator; and a policy integrator. The modules may be embodied as containerized services communicating over a message bus and persisting state to one or more data stores. The recited arrangement is illustrative; functions may be combined or distributed without departing from the invention.

1.6.2 2. Threat-intelligence ingestion module

The ingestion module continuously acquires external threat intelligence (for example, via STIX/TAXII, REST, CSV, syslog, and message-bus connectors from sources such as MISP, OTX, VirusTotal, and AbuseIPDB) and internal telemetry (for example, SIEM, IDS, EDR, and firewall logs). Acquired data is normalized to a canonical schema with attached provenance and timestamps; deduplicated by indicator value within a configurable time window with confidence merging; filtered by allow/deny lists and confidence thresholds; and enriched by extracting indicators of compromise (IP, domain, file hash), referenced vulnerabilities (CVEs), and mappings to a standardized adversary-technique taxonomy such as MITRE ATT&CK. Normalized records are published to downstream consumers and persisted for history and audit.

1.6.3 3. Artificial-intelligence decision engine

The decision engine scores incoming threats along severity, confidence, and urgency axes and detects trends (for example, a spike in a particular service or vulnerability in a particular region). Subject to the policy constraints of Section 9, it generates a *deployment instruction* specifying at least: a honeypot type; a target provider and region; an interaction level; an emulated vulnerability profile; a set of randomization parameters; and a time-to-live. Each deployment instruction is accompanied by a machine-generated rationale enumerating the top contributing signals, rendering every autonomous action explainable and auditable. In some embodiments the engine adapts its policy from observed outcomes using a multi-armed bandit or reinforcement-learning method in which the reward is a function of realized engagement depth and intelligence yield.

1.6.4 4. Honeypot orchestrator

The orchestrator validates a deployment instruction, rejecting unsupported or non-compliant plans, and provisions the corresponding infrastructure as one or more virtual machines or containers using infrastructure-as-code tooling and cloud application programming interfaces. It applies network controls (security groups, egress restrictions, optional reverse proxies and decoy domain names), performs health checks with automatic retry and rollback on failure, and maintains an inventory of instances and their state transitions, including time-to-live cleanup and recycling. Referring to FIG. 7, deployments may span multiple cloud providers, regions, and tenants.

1.6.5 5. Dynamic configuration manager and deception-surface randomization

The configuration manager installs and activates honeypot software stacks (for example, an SSH/Telnet decoy, a malware-capture decoy, or a web-application decoy), emulates services

on configured ports with realistic banners and responses, and injects vulnerability traits (weak credentials, outdated version strings, simulated vulnerable endpoints).

Referring to FIG. 3, the configuration manager further **randomizes the deception surface** of each deployment. In a representative embodiment, for each new decoy the manager samples, from constrained distributions, two or more of: (i) the operating- system fingerprint exposed to active scanners; (ii) the set and ordering of open ports; (iii) service banners and version identifiers; (iv) error and prompt strings; (v) response timing and jitter; (vi) filesystem and command-output artifacts; and (vii) the specific emulated vulnerabilities. The sampled parameters are constrained to remain mutually consistent (so the decoy remains believable) yet are varied across deployments such that successive decoys do not share an identical fingerprint. A registry of recently used fingerprints may be maintained so that the sampler actively diversifies away from prior deployments. This randomization defeats signature-based and consistency-based honeypot-detection reconnaissance, materially extending the useful life and intelligence yield of each decoy.

1.6.6 6. Behavioral-cognitive intent model

Referring to FIG. 4, a behavioral-cognitive intent model infers the probable intent of an interacting adversary from *early, pre-action* interaction features — that is, from behavior observable before the adversary commits a terminal action such as payload delivery. In a representative embodiment, for an interactive session the model computes features including protocol and client-version characteristics, authentication-attempt structure (count, distinctness, credential entropy, use of common usernames), session-establishment parameters, and timing, deliberately excluding post-action features to avoid leakage and to enable *early* inference. A trained classifier maps these features to (a) an engagement likelihood and (b) an intent class or attacker archetype (for example, brute-force, tunneling, execution, or malware-delivery oriented). The decision engine and configuration manager consume the inferred intent to (i) adapt the configuration of the currently engaged decoy in real time toward the inferred objective so as to deepen engagement, and (ii) bias the placement and configuration of subsequent decoys toward the prevailing intent distribution. In further embodiments, sessions are clustered into behavioral archetypes that parameterize future deployments. This early-intent capability distinguishes the invention from reactive deception that adapts only after an adversary has acted.

1.6.7 7. Constrained large-language-model interaction

In some embodiments the configuration manager incorporates a constrained large-language-model component that generates human-like interactive responses to adversary commands (for example, in an SSH session), increasing believability and engagement depth. The component operates under safety guardrails — including output filtering, capability restriction, and policy checks — that prevent the decoy from performing or facilitating actual malicious activity, exfiltrating real data, or emitting unsafe content. The guardrails are enforced independently of the language model, so that a model error cannot by itself breach the safety boundary.

1.6.8 8. Real-time monitoring, alerting, and correlation

The monitoring subsystem ingests session and network telemetry from deployed decoys; applies both rule-based detection (for example, known commands, file upload, reverse-shell patterns) and machine-learning anomaly detection for novel TTPs and unusual command sequences; and generates enriched alerts annotated with geographic origin, vulnerability linkage, adversary-technique

mapping, and deployment context. The threat-intelligence logger normalizes and persists logs, sessions, alerts, and decisions; correlates interactions into longitudinal attacker profiles by stitching sessions and grouping campaigns over time; computes analytics (top indicators, targeted vulnerabilities, geographic distribution, dwell time); and exposes query and export interfaces, subject to encryption, retention, and access governance.

1.6.9 9. Policy integrator and guardrails

Referring to FIG. 5, a policy integrator constrains all autonomous behavior at both decision time (within the decision engine) and execution time (within the orchestrator). Enforced guardrails include: permitted clouds, regions, and geographies with deny-lists; permitted honeypot templates and interaction levels per tenant; budget and quota constraints (maximum instances, maximum spend, maximum public exposure); network-exposure rules (ports, source-address allow-lists, proxy/WAF requirements); data-retention and personally-identifiable-information handling constraints; and optional human approval for high-risk deployments. The decision engine cannot emit, and the orchestrator will not execute, an instruction that violates an applicable policy; rejected instructions are logged with reasons.

1.6.10 10. Closed feedback loop

Referring to FIG. 2, the modules form a closed loop: threat intelligence is ingested and normalized; the decision engine emits a policy-constrained, explainable deployment instruction; the orchestrator provisions and the configuration manager configures and randomizes a decoy; the monitoring subsystem and intent model observe interaction, infer intent, and produce enriched alerts and attacker profiles; and observed behavior and newly extracted indicators are returned to the decision engine to refine subsequent strategy. The deception population thereby continuously adapts to the prevailing threat environment and to inferred adversary intent.

1.6.11 11. Data model

Referring to FIG. 6, principal domain objects include: a ThreatIntellItem (normalized indicator with source, type, value, confidence, severity, and first/last-seen); a DeploymentInstruction (type, provider, region, interaction level, vulnerability profile, randomization parameters, TTL, rationale); a HoneypotInstance (identifier, address, ports, template, region, state, TTL, tenant); a SessionRecord (session identifier, decoy identifier, source, commands, files, captures, timestamps, extracted behavioral features); an Alert (identifier, severity, confidence, technique mapping, context, recommended actions); an AttackerProfile (fingerprints, infrastructure, tooling, recurring indicators, campaign linkage, inferred intent); and a Policy (limits, allow/deny lists, retention, approval flags).

1.6.12 12. Alternative embodiments

The invention is not limited to the recited embodiments. The deception assets may include low-, medium-, and high-interaction decoys, decoy tokens, and decoy credentials; the computing environments may be public cloud, private cloud, on-premises, or operational-technology networks; the intent model may be embodied as a gradient-boosted, random-forest, or neural classifier; the language-model component is optional; and the randomization, intent-inference, and policy subsystems may each be practiced independently. Functions described as separate modules may be combined, and any single module may be distributed across hosts.

1.6.13 13. Example operational scenario

Threat intelligence indicates a surge in SSH brute-force activity targeting a particular region. The decision engine, within budget and region policy, emits an explainable instruction to deploy a medium-interaction SSH decoy in that region with a freshly randomized fingerprint and weak-credential profile, with a 72-hour TTL. The orchestrator provisions a container; the configuration manager randomizes the banner, port artifacts, and timing, and activates the SSH stack. An adversary connects; the intent model, from early authentication structure and client characteristics, infers a tunneling rather than brute-force objective and a high engagement likelihood; the configuration manager adapts the decoy to present a plausible pivot path, deepening engagement, while the decision engine biases subsequent deployments toward tunneling-oriented decoys. The monitoring subsystem emits a technique-mapped alert and the logger links the session to an existing campaign profile. The observed behavior updates the decision policy, improving future placement.

1.7 Claims

What is claimed is:

1. A system for adaptive cyber-deception, comprising one or more processors and memory storing instructions that, when executed, cause the system to: (a) ingest threat-intelligence data from one or more external feeds and internal telemetry sources and normalize the ingested data to a canonical schema; (b) generate, by an artificial-intelligence decision engine and subject to one or more machine-enforceable policies, a deployment instruction specifying at least a deception-asset type, a deployment location, an interaction level, and an emulated vulnerability profile, the deployment instruction being accompanied by a machine-generated rationale identifying contributing signals; (c) provision, by an orchestrator, at least one deception asset in accordance with the deployment instruction across one or more computing environments; (d) configure the at least one deception asset by a configuration manager that randomizes a deception surface of the deception asset such that successive deployments do not present an identical fingerprint; (e) monitor interaction of an adversary with the at least one deception asset and generate an enriched alert characterizing the interaction; and (f) return data derived from the monitored interaction to the artificial-intelligence decision engine to adapt a subsequent deployment instruction in a closed feedback loop.
2. The system of claim 1, wherein randomizing the deception surface comprises sampling, from constrained distributions and subject to a mutual-consistency constraint, two or more of: an operating-system fingerprint, a set of open ports, a service banner, a version identifier, an error or prompt string, a response timing, a filesystem artifact, and an emulated vulnerability.
3. The system of claim 2, wherein the instructions further cause the system to maintain a registry of previously deployed fingerprints and to bias the sampling away from fingerprints in the registry.
4. The system of claim 1, wherein the one or more machine-enforceable policies are applied both at a time the deployment instruction is generated and at a time the deception asset is provisioned, and constrain at least one of: a permitted cloud region, a permitted template, a budget or quota, a maximum public exposure, a network-exposure rule, and a data-retention rule.
5. The system of claim 1, wherein the instructions further cause the system to infer, by a behavioral-cognitive intent model and from interaction features observed before the adversary commits a terminal action, an intent class or attacker archetype, and to adapt, based on the inferred intent class or attacker archetype, at least one of: a configuration of an engaged deception asset and a placement of a subsequent deception asset.

6. The system of claim 5, wherein the interaction features comprise authentication-attempt structure and client characteristics and exclude post-action features.
7. The system of claim 1, wherein the configuration manager comprises a constrained large-language-model component configured to generate human-like interactive responses to adversary commands, the responses being bounded by safety guardrails, enforced independently of the large-language-model component, that prevent the deception asset from facilitating actual malicious activity.
8. The system of claim 1, wherein the artificial-intelligence decision engine adapts the subsequent deployment instruction using a reinforcement-learning or multi-armed- bandit policy whose reward is a function of an engagement depth or an intelligence yield realized from the monitored interaction.
9. The system of claim 1, wherein the instructions further cause the system to correlate a plurality of monitored interactions across time into an attacker profile linking recurring indicators, tooling, and infrastructure into a campaign, and to annotate the enriched alert with a mapping to a standardized adversary-technique taxonomy.
10. The system of claim 1, wherein the deployment location is selected from a plurality of cloud providers or regions, and wherein each provisioned deception asset is assigned a time-to-live after which the orchestrator recycles the deception asset.
11. A method for adaptive cyber-deception, comprising: ingesting and normalizing threat-intelligence data; generating, by an artificial-intelligence decision engine and subject to one or more machine-enforceable policies, a deployment instruction specifying a deception-asset type, a deployment location, an interaction level, and an emulated vulnerability profile; provisioning at least one deception asset in accordance with the deployment instruction; configuring the at least one deception asset with a randomized deception surface such that successive deployments do not present an identical fingerprint; monitoring adversary interaction with the at least one deception asset and generating an enriched alert; and adapting a subsequent deployment instruction based on data derived from the monitored interaction in a closed feedback loop.
12. The method of claim 11, further comprising emitting, with the deployment instruction, a machine-generated rationale identifying the threat-intelligence signals that contributed to the deployment instruction, thereby rendering the autonomous deployment decision auditable.
13. The method of claim 11, further comprising rejecting, at provisioning time, any deployment instruction that violates an applicable policy constraint, and logging the rejection with a reason.
14. The method of claim 11, further comprising inferring an attacker intent from interaction features observed before the adversary commits a terminal action, and adapting a configuration of an engaged deception asset toward the inferred intent so as to deepen engagement.
15. A method for resisting reconnaissance of deception assets, comprising, for each of a plurality of deception assets to be deployed: sampling, from constrained distributions and subject to a mutual-consistency constraint, values for a plurality of fingerprint dimensions including at least an operating-system fingerprint, an open-port set, and a service banner; biasing the sampling away from fingerprints recorded in a registry of previously deployed deception assets; and configuring the deception asset with the sampled values such that no two of the plurality of deception assets present an identical fingerprint while each remains internally consistent.
16. The method of claim 15, further comprising recording the sampled fingerprint in the registry, and varying the fingerprint of a given deployment over its lifetime.

17. A method for intent-driven cyber-deception, comprising: receiving interaction telemetry from a deception asset; computing, from the telemetry, behavioral features observable before an interacting adversary commits a terminal action; inferring, by a trained model and from the behavioral features, an engagement likelihood and an intent class; and adapting, based on the inferred intent class, at least one of a configuration of the deception asset and a deployment of a subsequent deception asset.
18. The method of claim 17, wherein the behavioral features comprise an authentication-attempt count, a count of distinct credentials, a credential-length or entropy statistic, a protocol or client-version characteristic, and a timing characteristic, and exclude command-execution and payload features.
19. The method of claim 17, further comprising clustering a plurality of sessions into attacker archetypes and parameterizing subsequent deployments according to a prevailing archetype distribution.
20. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform the method of claim 11.
21. The non-transitory computer-readable medium of claim 20, wherein the instructions further cause the processors to perform the randomization of claim 15 and the intent inference of claim 17.
22. The system of claim 1, wherein the computing environments comprise two or more of a public cloud, a private cloud, an on-premises network, and an operational-technology network, and wherein the system supports multiple isolated tenants each governed by tenant-specific policy.
23. The system of claim 1, wherein the enriched alert is routed to one or more of a security-information-and-event-management system, a security-orchestration system, and a notification channel, and wherein indicators extracted from the interaction are fed back to the threat-intelligence ingestion module.
24. The system of claim 5, wherein the inferred intent class is one of a brute-force, a tunneling, an execution, and a malware-delivery class, and wherein the configuration manager presents, responsive to the inferred class, a decoy characteristic selected to deepen engagement for that class.
25. The method of claim 11, wherein the deception-asset type is selected from a low-interaction decoy, a medium-interaction decoy, a high-interaction decoy, a decoy token, and a decoy credential.

1.8 Abstract

A closed-loop, policy-bounded, artificial-intelligence system autonomously deploys and configures deception assets based on live threat intelligence and inferred attacker intent. A decision engine ingests and normalizes threat intelligence and, subject to machine-enforceable cost, region, exposure, and compliance policy, emits an explainable instruction specifying a decoy type, location, interaction level, emulated vulnerability profile, randomization parameters, and time-to-live. An orchestrator provisions the decoy across one or more environments, and a configuration manager randomizes its deception surface — operating-system fingerprint, ports, banners, timing, and emulated vulnerabilities — so successive deployments share no fingerprint and resist reconnaissance. A behavioral-cognitive intent model infers, from early pre-action interaction features, an engagement likelihood and an attacker archetype, and adapts the engaged decoy and subsequent placements accordingly; an optional constrained language-model component deepens interactive believability under independent safety guardrails. A monitoring and correlation subsystem turns interaction

into technique-mapped alerts and longitudinal attacker profiles and returns observed behavior to the decision engine, continuously adapting the deception population to the prevailing threat environment.